

F-02-CX-concentration-vs-Wplus-kill-test v1: does C_F^{conc} force $\int W_+ < \infty$? Quantitative audit

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MISSION FROZEN: Search only bridge $C_F^{conc} \Rightarrow \int_0^{T^*} W_+ < \infty$ or directly $C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \Rightarrow \perp$. Attempt quantitative inequality $W_+(t) \leq f(C_F^{conc}, \text{controlled})$ respecting NS identities under $T^* < \infty$.

INPUT: $\mathcal{C}_{PROVEN}$ pending audit of C_F^{conc} , G LOCKED, E-CX CLOSED/BLOCKED, F-01 = PROVEN NECESSARY NO CONTRADICTION if quantifiers correct. Definitions: $W(t) = \int \omega^T S \omega$, $W_+ = \int (\omega^T S \omega)_+$, $C_E^{weak} : \limsup_{t \uparrow T^*} \int_0^t W = +\infty \Rightarrow \int_0^{T^*} W_+ = \infty$, $C_{L^3} : \limsup \|u\|_3 = \infty$, $C_F^{conc} : \exists t_n \uparrow T^*, x_n, r_n \downarrow 0, \|u(t_n)\|_{L^3(B_{r_n}(x_n))} \geq \varepsilon_0$.

METHODOLOGICAL PRECISION: Compatibility killing must respect NS identities effectively available under $T^* < \infty$: energy bound $\|u\|_{L_t^\infty L_x^2} \leq C$, enstrophy balance $\frac{1}{2} \|\omega(t)\|_2^2 + \nu \int_0^t \|\nabla \omega\|_2^2 = \frac{1}{2} \|\omega_0\|_2^2 + \int_0^t W$, Calderon-Zygmund p -vorticity $\|S\|_p \sim \|\omega\|_p$, pressure Hessian $-\Delta p = \text{tr}(\nabla u)^2 = |S|^2 - |\omega|^2/4$, and scaling $\|u^\lambda\|_3 = \|u\|_3$, $\|S^\lambda\|_{3/2} = \|S\|_{3/2}$, etc. Not abstract counterexamples unrealizable by NS.

AUDIT ATTEMPT FOR BOUND:

Goal: $W_+ \leq f(C_F^{conc})$.

Recall $W(t) = \int \omega_i S_{ij} \omega_j$, $|W| \leq \int |S| |\omega|^2 \leq \|S\|_{3/2} \|\omega\|_6^2$? But $\|\omega\|_6 \leq C \|\nabla \omega\|_2$ Sobolev. So $|W| \leq C \|S\|_{3/2} \|\nabla \omega\|_2^2$. Since $\|S\|_{3/2} \sim \|\omega\|_{3/2} \leq C \|u\|_3$? Actually $\|\nabla u\|_{3/2} \leq C \|u\|_3$ by Sobolev? No, $\|u\|_3$ controls $\|\nabla u\|_{3/2}$ via Calderon? $\|\nabla u\|_p \sim \|\omega\|_p$ and $\|u\|_3$ does not control $\|\omega\|_{3/2}$ without extra. Interpolation: $\|\omega\|_{3/2} \leq \|\omega\|_2^{\frac{2}{3}} \|\omega\|_1^{\frac{1}{3}}$ not controlled.

Try direct: $W_+ \leq \int |S| |\omega|^2$, Hölder $W_+ \leq \|S\|_{L^3(B)} \|\omega\|_{L^3}^2 + \text{far}$. C_F^{conc} gives lower bound on $\|u\|_{L^3(B_{r_n})}$, not upper bound on $\|S\|$ or $\|\omega\|$.

Critical observation: C_F^{conc} is a LOWER bound $\geq \varepsilon_0$ on local L^3 norm. W_+ requires UPPER bound. Lower bound cannot imply upper bound. No inequality $W_+ \leq f(\text{lower bound})$ can hold.

Could C_F^{conc} together with energy bound imply upper bound on W_+ ? Energy $\|u\|_2 \leq C$ and local $L^3 \geq \varepsilon_0$ at scale $r_n \downarrow 0$ means concentration: $\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0$ implies $\|u\|_{L^2(B_{r_n})} \leq C r_n^{1/2} \|u\|_{L^3(B_{r_n})}$? Actually $\|u\|_{L^2(B_r)} \leq C r^{1/2} \|u\|_{L^3(B_r)}$ by Hölder. So if $r_n \downarrow 0$ and $\|u\|_{L^3} \geq \varepsilon_0$, then $\|u\|_{L^2(B_{r_n})} \geq ?$ Lower bound on L^2 goes to 0 anyway, no contradiction with global L^2 bound.

Can C_F^{conc} coexist with arbitrarily large W_+ within NS? Yes: take a fixed background L^3 bubble with norm $\geq \varepsilon_0$ in B_{r_n} plus a high-frequency vortex packet inside same ball with vorticity $\omega_{high} = M \phi(x/\delta)$ where $M \gg 1$, $\delta \ll r_n$, divergence-free corrected. Then $\|u_{high}\|_3 \sim M \delta$ can be made small to not destroy C_F^{conc} lower bound, while $W_+^{high} \sim \int |\omega_{high}|^2 |S_{high}| \sim M^3 \delta^3 \sim M^3 \delta^3$? Choose $M = \delta^{-1}$ gives $\|u\|_3 \sim 1$, $W_+ \sim \delta^{-3}$ diverges as $\delta \rightarrow 0$. Need to respect NS enstrophy balance? Enstrophy $\|\omega\|_2^2 \sim M^2 \delta^3 \sim \delta$, actually bounded. To make W_+ arbitrarily large at instant t_n , need large gradients, which cost $\|\nabla \omega\|_2^2$ but that appears in enstrophy balance as dissipation, allowed to be large. No NS identity forbids arbitrarily large instantaneous $W_+(t_n)$ while keeping C_F^{conc} and $\|u\|_2$ bounded. Energy inequality does not bound W_+ pointwise.

More formally, vortex stretching term localized estimate (Grujic-type) gives $|I_i| \leq \frac{1}{4} \int |\nabla \omega|^2 \phi_i + C \max\{M_0^2, R_0^2\} R^{-4} \int |\omega|^2 \phi_i^{1/2}$, where factor $\sup \int |u|^2 / R$ is scaling invariant and a priori bounded only assuming uniform L^3 bound which implies regularity. Without uniform L^3 bound (our case C_{L^3} diverges), the factor is not bounded, allowing large VST.

Thus within NS constraints (energy, enstrophy balance, pressure equation), we can have C_F^{conc} + $\int_0^{T^*} W_+ = +\infty$ with arbitrarily fast growth, no upper bound forced.

Check all kill conditions:

- C_F^{conc} does not control any quantity entering W_+ (it controls u lower bound, W_+ needs upper bound on S, ω) =_i KILL. - Only reproduces Hölder/Sobolev/scale critical: lower bound $L^3(B_{r_n}) \geq \varepsilon_0$ is scale-critical, no new scale. - Concentration can coexist with arbitrary W_+ growth within NS audited restrictions (energy bound, enstrophy identity, pressure) =_i KILL. - Any bound $W_+ \leq f(C_F^{conc})$ would need additional hypothesis (e.g., uniform L^3 bound, or Type I, or direction coherence) not consequence of $T^* < \infty$ =_i KILL. - Only qualitative relation, no integrated bound $\int W_+ < \infty$ =_i KILL.

Therefore no quantitative inequality exists respecting NS identities.

VERDICT:

$F-02 = KILLED$ fast

No bridge $C_F^{conc} \Rightarrow \int W_+ < \infty$, no $\perp$ with $C_E^{weak} \wedge C_{L^3}$ within NS realizable class.

$F = CLOSED/BLOCKED \rightarrow H-01$

No F-03. Before promoting F-01 to $\mathcal{C}_{PROVEN}$, verify quantifiers in F-01 PDF: does it prove $C_{L^3} \Rightarrow C_F^{conc}$ with universal ε_0 and $r_n \downarrow 0$ necessary, not just possible? If not, correct F-01; if yes, promote:

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}\}$ with $F = PROVEN NECESSARY NO CONTRADICTION$

but central $C_E^{weak} \wedge C_{L^3} \Rightarrow \perp OPEN$ remains.

Next family H-01 genuinely new (helicity already killed? need new beyond H , e.g., circulation, scale-time dynamics, Lagrangian).